Deformation-Stable Ordering of CP-Phase Admission in Constrained Integer Carriers

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We define a constrained discrete phase carrier $K_\Lambda \subset \mathbb{Z}_N$ generated by weighted integer combinations modulo N subject to primitivity and a bounded weighted length Λ. The carrier coverage $|K_\Lambda|/N$ grows monotonically with Λ and saturates at a finite cutoff $\Lambda_{\rm sat}$. For a target CP-violating phase δ, we define the admission cutoff $\Lambda_{\rm adm}$ as the smallest Λ for which the minimal circular phase distance satisfies $d_{\min}(\Lambda;\delta) \le \Delta$. The dimensionless ratio $\eta = \Lambda_{\rm adm}/\Lambda_{\rm sat}$ enables scale-free comparison across different N and across carrier deformations. Applying this construction uniformly to the CKM phase and to representative PMNS phases (normal and inverted ordering), we find a deformation-stable ordering $\eta_{\rm CKM} < \eta^{\rm IO}_{\rm PMNS} < \eta^{\rm NO}_{\rm PMNS}$, with all admissions occurring well below saturation. This hierarchy indicates a structural ordering in admission depth independent of carrier scale.

I. INTRODUCTION

CP violation in the Standard Model is encoded in rephasing-invariant phases of the quark and lepton mixing matrices. While the quark-sector CP phase is precisely measured, the leptonic CP phase remains less constrained and depends on the neutrino mass ordering. Rather than proposing a dynamical origin for these phases, this work addresses a structural question: when identical discrete constraints are imposed, how deeply must a carrier be populated before each physical CP phase is admitted?

II. DISCRETE PHASE CARRIER

We consider discrete phases

$$\theta_k = \frac{2\pi k}{N}, \qquad k \in \mathbb{Z}_N, \tag{1}$$

and generate admissible residues via

$$k \equiv w_1 a + w_2 b + w_3 c \pmod N, \tag{2}$$

subject to

$$w_1|a| + w_2|b| + w_3|c| \le \Lambda, \tag{3}$$
$$\gcd(a,b,c) = 1, \tag{4}$$
$$(a,b,c) \ne (0,0,0). \tag{5}$$

For fixed $(N, \mathbf{w})$ and cutoff Λ, the admissible residue set $K_\Lambda \subset \mathbb{Z}_N$ defines the carrier coverage

$$C(\Lambda) = \frac{|K_\Lambda|}{N}. \tag{6}$$

As Λ increases, $C(\Lambda)$ grows monotonically until saturation.

III. ADMISSION METRIC AND NORMALIZED DEPTH

Agreement with a target phase δ is measured using the circular distance on S^1,

$$d(\theta,\delta) = \min_{n\in\mathbb{Z}} |\theta - \delta + 2\pi n|. \tag{7}$$

For a given carrier K_Λ, define

$$d_{\min}(\Lambda;\delta) = \min_{k\in K_\Lambda} d(\theta_k,\delta). \tag{8}$$

The *admission cutoff* $\Lambda_{\rm adm}$ is the smallest Λ such that

$$d_{\min}(\Lambda;\delta) \le \Delta, \tag{9}$$

for a fixed tolerance Δ.

To enable comparison across different N and scaffold choices, we define the dimensionless *normalized admission depth*

$$\eta = \frac{\Lambda_{\rm adm}}{\Lambda_{\rm sat}}, \tag{10}$$

where $\Lambda_{\rm sat}$ is the smallest cutoff for which $C(\Lambda) = 1$.

A. Computation

For fixed $(N, \mathbf{w})$, we enumerate all integer triples $(a,b,c) \in \mathbb{Z}^3$ satisfying $\gcd(a,b,c) = 1$ and $w_1|a|+w_2|b|+w_3|c| \le \Lambda$, collecting residues $k \equiv w_1 a + w_2 b + w_3 c \pmod N$ to form K_Λ. The saturation cutoff $\Lambda_{\rm sat}$ is the smallest Λ for which $|K_\Lambda| = N$. For a target phase δ, the admission cutoff $\Lambda_{\rm adm}$ is the smallest Λ satisfying the distance criterion above. All results are deterministic given $(N, \mathbf{w}, \delta, \Delta)$.

TABLE I. Representative normalized admission depths for $N = 120$. The complete table across all tested $(N, \mathbf{w})$ is provided in Appendix A.

Weights	$\eta_{\rm CKM}$	$\eta^{\rm IO}_{\rm PMNS}$	$\eta^{\rm NO}_{\rm PMNS}$
8-15-24	0.328	0.198	0.483
7-15-24	0.244	0.244	0.633
8-14-25	0.220	0.220	0.570
9-15-23	0.667	0.225	0.549

IV. INPUTS AND DEFORMATION SET

We use the central values

$$\delta_{\rm CKM} = 65.718^\circ, \tag{11}$$
$$\delta^{\rm NO}_{\rm PMNS} = 177^\circ, \tag{12}$$
$$\delta^{\rm IO}_{\rm PMNS} = 285^\circ, \tag{13}$$

with tolerances $\Delta_{\rm CKM} = 1.5^\circ$ and $\Delta_{\rm PMNS} = 10^\circ$.

To test deformation stability we evaluate the integer scaffolds

$$(8, 15, 24), \quad (7, 15, 24), \quad (8, 14, 25), \quad (9, 15, 23), \tag{14}$$

for representative values $N \in \{60, 100, 120, 180\}$.

V. RESULTS

a. Result. Across all tested $(N, \mathbf{w})$, we observe a strict ordering in normalized admission depth,

$$\eta_{\rm CKM} < \eta^{\rm IO}_{\rm PMNS} < \eta^{\rm NO}_{\rm PMNS}. \tag{15}$$

Because η is defined relative to the saturation scale $\Lambda_{\rm sat}$, this ordering is insensitive to absolute carrier scale and persists under scaffold deformations that modify coverage growth.

VI. CONCLUSION AND FALSIFIABILITY

We identify a deformation-stable, scale-invariant ordering governing the admission of CP-violating phases into constrained discrete carriers. Under identical structural constraints, the CKM phase is admitted at lower normalized depth than representative leptonic phases, with the PMNS normal-ordering target requiring the deepest carrier penetration.

This framework is falsifiable. Observation of a leptonic CP phase that admits at lower normalized depth than the CKM phase under the same carrier constraints would invalidate the ordering identified here.

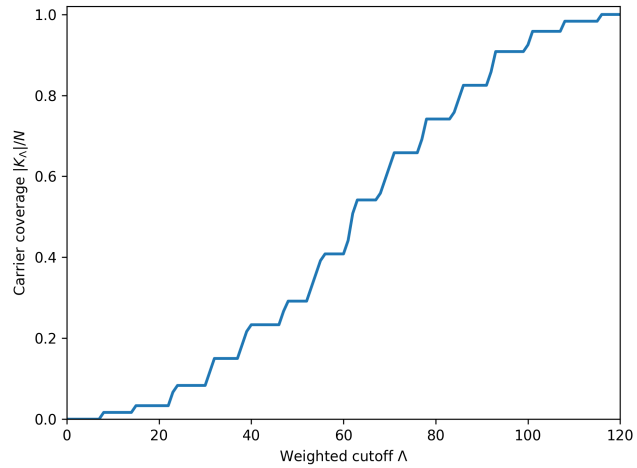

FIG. 1. Carrier coverage $|K_\Lambda|/N$ versus weighted cutoff Λ for $N = 120$ (baseline scaffold $\mathbf{w} = (8, 15, 24)$). Admission of the CKM and PMNS target phases occurs well below saturation, indicating that the observed ordering is not a saturation artifact.

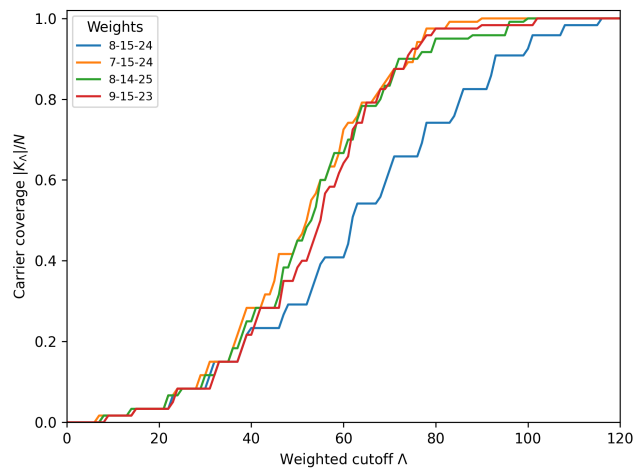

FIG. 2. Coverage curves for $N = 120$ under small scaffold deformations. Similar growth and comparable saturation scales support robustness of η as a comparative invariant.

Appendix A: Full deformation table

TABLE II: Admission cutoff $\Lambda_{\rm adm}$, saturation cutoff $\Lambda_{\rm sat}$, and normalized depth η across all tested $(N, \mathbf{w})$.

Weights	N	Sector	$\Lambda_{\rm adm}$	$\Lambda_{\rm sat}$	η
8-15-24	60	CKM	71	86	0.826
8-15-24	60	PMNS_NO	31	86	0.360
8-15-24	60	PMNS_IO	47	86	0.547
8-15-24	100	CKM	78	110	0.709
8-15-24	100	PMNS_NO	47	110	0.427
8-15-24	100	PMNS_IO	23	110	0.209
8-15-24	120	CKM	38	116	0.328
8-15-24	120	PMNS_NO	56	116	0.483
8-15-24	120	PMNS_IO	23	116	0.198
8-15-24	180	CKM	63	138	0.457
8-15-24	180	PMNS_NO	84	138	0.609
8-15-24	180	PMNS_IO	38	138	0.275

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Weights	N	Sector	Λ_{adm}	Λ_{sat}	η
7-15-24	60	CKM	59	64	0.922
7-15-24	60	PMNS_NO	29	64	0.453
7-15-24	60	PMNS_IO	43	64	0.672
7-15-24	100	CKM	60	90	0.667
7-15-24	100	PMNS_NO	50	90	0.556
7-15-24	100	PMNS_IO	22	90	0.244
7-15-24	120	CKM	22	90	0.244
7-15-24	120	PMNS_NO	57	90	0.633
7-15-24	120	PMNS_IO	22	90	0.244
7-15-24	180	CKM	63	102	0.618
7-15-24	180	PMNS_NO	84	102	0.824
7-15-24	180	PMNS_IO	36	102	0.353
8-14-25	60	CKM	39	88	0.443
8-14-25	60	PMNS_NO	30	88	0.341
8-14-25	60	PMNS_IO	14	88	0.159
8-14-25	100	CKM	46	100	0.460
8-14-25	100	PMNS_NO	47	100	0.470
8-14-25	100	PMNS_IO	22	100	0.220
8-14-25	120	CKM	22	100	0.220
8-14-25	120	PMNS_NO	57	100	0.570
8-14-25	120	PMNS_IO	22	100	0.220
8-14-25	180	CKM	33	104	0.317
8-14-25	180	PMNS_NO	84	104	0.808
8-14-25	180	PMNS_IO	33	104	0.317
9-15-23	60	CKM	65	65	1.000
9-15-23	60	PMNS_NO	32	65	0.492
9-15-23	60	PMNS_IO	32	65	0.492
9-15-23	100	CKM	100	100	1.000
9-15-23	100	PMNS_NO	47	100	0.470
9-15-23	100	PMNS_IO	23	100	0.230
9-15-23	120	CKM	68	102	0.667
9-15-23	120	PMNS_NO	56	102	0.549
9-15-23	120	PMNS_IO	23	102	0.225
9-15-23	180	CKM	33	110	0.300
9-15-23	180	PMNS_NO	84	110	0.764
9-15-23	180	PMNS_IO	33	110	0.300